SYSADMIN & SECURITY HANDBOOK | SYSCOHADA REVISED (AUDCIF) | IFRS STANDARDS




*Figure 1: LedgerCore Health Dual-Standard Architecture — Unified Double-Entry Core connecting SYSCOHADA Revised (OHADA), IFRS Presentation, Biomedical Asset Lifecycle, and Hospital Cash Desks.*

## 1. System Deployment & Environment Setup

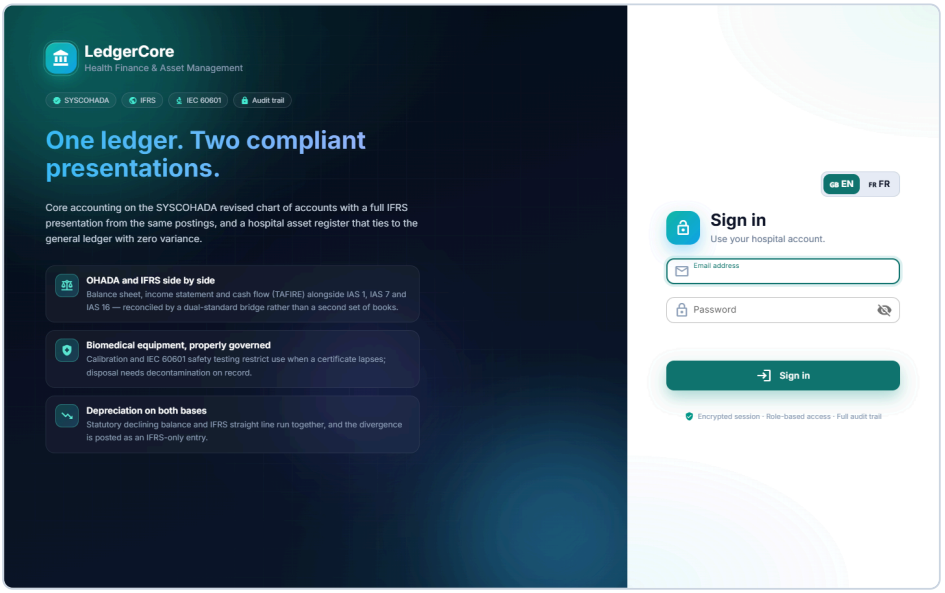



*Figure 2: Authentication Portal — Language Selection, Encrypted Session Handshake, and RBAC Profiles.*

LedgerCore Health operates as a high-performance decoupled architecture: - ⚡ **Backend API:** Node.js (Express, ES Modules, MySQL2/MariaDB connection pool) on port `4700`. - 🌐 **Frontend UI:** Python 3.10+ (NiceGUI, Quasar/Vue3, TailwindCSS) on port `8700`. - 🗄️ **Database:** MariaDB 10.5+ / MySQL 8.0 on port `3306` (Database: `ledgercore`).

### Step-by-Step Procedure: Starting Services on Windows Server

1. 🗄️ **Database Service:** Ensure MariaDB / MySQL is running via Windows Services or XAMPP Control Panel.
2. ⚡ **Launch Backend API:**

```
cd "C:\Core Accounting and asset\backend"
npm start
```

*Health Check Verification:* Open `http://127.0.0.1:4700/api/health` — must return status `healthy`.

3. 🌐 **Launch Frontend Service:**

```
cd "C:\Core Accounting and asset\frontend"
python main.py
```

*UI Access:* Open `http://127.0.0.1:8700` in Google Chrome or Microsoft Edge.

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## 2. Step-by-Step Procedure: User Provisioning & Granular RBAC Permissions

Under **Administration → Users & Permissions** ( `/users` ): 1. 👤 Click **+ Add User**. 2. 📝 Enter email, full name, phone number, and default language ( `en` or `fr` ). 3. 🔑 Set the initial temporary password and enable **Force password change on first sign-in**. 4. 🛡️ Assign one or multiple role profiles: - `admin` : System configuration, organization settings, user management. - `cfo` : Full financial posting, period close, depreciation, and reporting. - `accountant` : Voucher preparation, open-item matching, bank reconciliation. - `asset_manager` : Asset register, maintenance scheduling, physical inventory. - `biomed_tech` : Work order execution, IEC 60601 testing, calibration logging. - `services_tech` : Errands, shopping lists, vehicle fleet logs. - `ward_head` : Departmental asset view and breakdown ticket submission. 5. 💾 Click **Save User**.

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## 3. Step-by-Step Procedure: Organization Settings & Statutory Posting Rules

Under **Administration → Organization Settings** ( `/settings` ): - 💱 **Functional Currency:** Default `XAF` (Central African CFA franc) or `XOF` . - 📒 **Accounting Frameworks:** Select `OHADA` , `IFRS` , or `Both` (Dual-Standard Mode). - 🔢 **Posting Length (Account Code Digits):** Set statutory account digits (4, 6, or 8 digits). Note: Once transactions exist, posting length is permanently locked to preserve historical audit integrity. - ⏳ **Suspense / Clearing Account:** Default `471100` for unclassified banking receipts awaiting allocation.

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# 4. Cryptographic Seal Verification & Forensic Audit Trail




*Figure 3: 12-Step Month-End Closing Procedure & Cryptographic SHA-256 Hash Chaining in LedgerCore Health.*

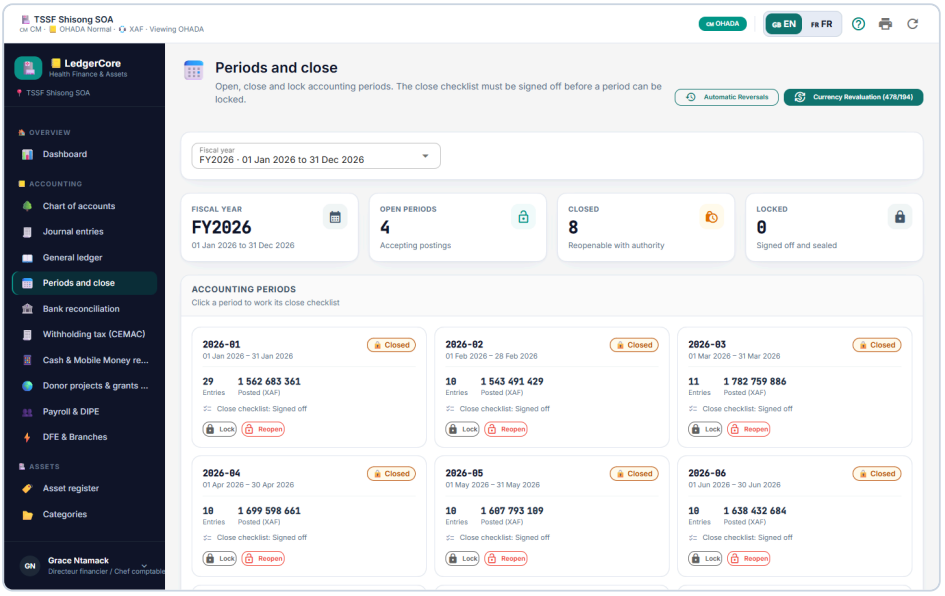



*Figure 4: Period Management Screen — Verifying Audit Seal Integrity and Permanent Chained Hashes.*

LedgerCore incorporates an immutable **SHA-256 blockchain-style seal** on journal entries: - 🔗 Every posted entry hashes the previous entry's hash, timestamp, user ID, account lines, and monetary amounts. - 🛡️ Under **Reporting → Audit Seal Verification**, run the integrity scanner: - If any row in the `journal_entries` or `journal_lines` database table was altered outside the application (e.g., direct SQL injection or manual database update), the cryptographic verification fails immediately with the exact tampered row number highlighted in red.

## 5. Automated Database Backup & Disaster Recovery

### Automated Daily Dump (PowerShell Script):

```
$Date = Get-Date -Format "yyyy-MM-dd_HH-mm"
$BackupFile = "C:\Backups\ledgercore_backup_$Date.sql"
mysqldump.exe -u root -p"your_password" --single-transaction --quick --routines --triggers ledg
```

### Restoration Protocol:

```
mysql.exe -u root -p"your_password" ledgercore < "C:\Backups\ledgercore_backup_2026-10-18.sql"
```